Imperfect Competition in Markets for Short-Circuit Current Services

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Faults in electric power systems

Unintentional **short circuits**:

- Wear-and-tear (transformers, cable insulation)
- Lightning
- Tree contact
- Wildlife…



General strategy: **de-energize fault** by opening circuit

Current is measured in real time, and **circuit breakers trip** when **current exceeds** a pre-defined **threshold**

Short-Circuit Current decline and consequences

- **Inverter-based Resources** (IBR) contribute much less Short-Circuit Current (SCC) than Synchronous Generators (SGs)

IBR (**1–3 p.u.** SCC)



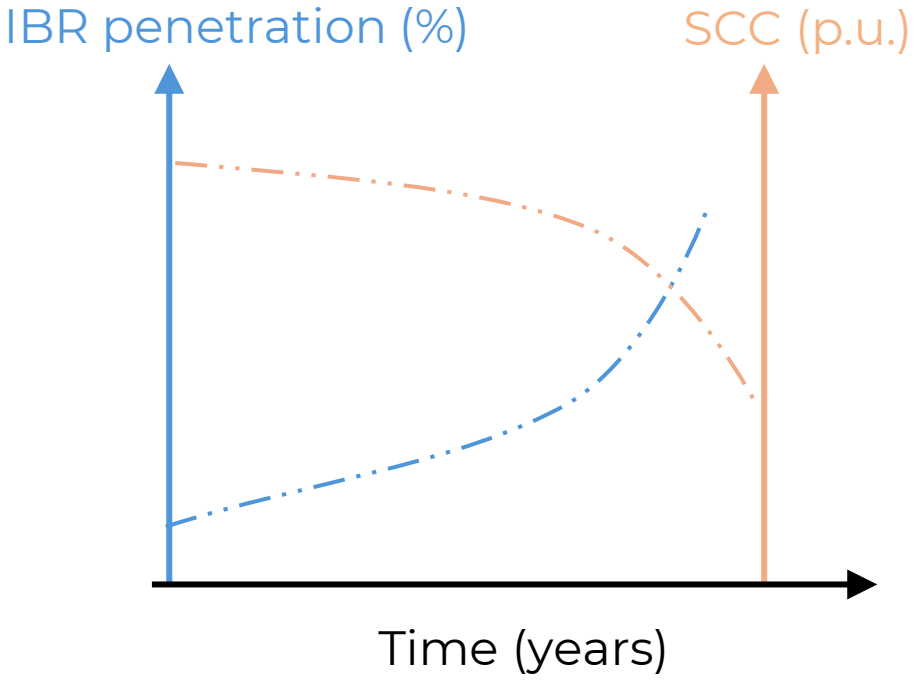


Electronic converters (**burning of transistors**)

SGs (**5–8 p.u.** SCC)



Electromechanical physics



Short-Circuit Current decline and consequences

- **Inverter-based Resources** (IBR) contribute much less Short-Circuit Current (SCC) than Synchronous Generators (SGs)

If fault current is reduced, the relay may **not be triggered**, so the fault ***would not*** be cleared

IBR (**1–3 p.u.** SCC)



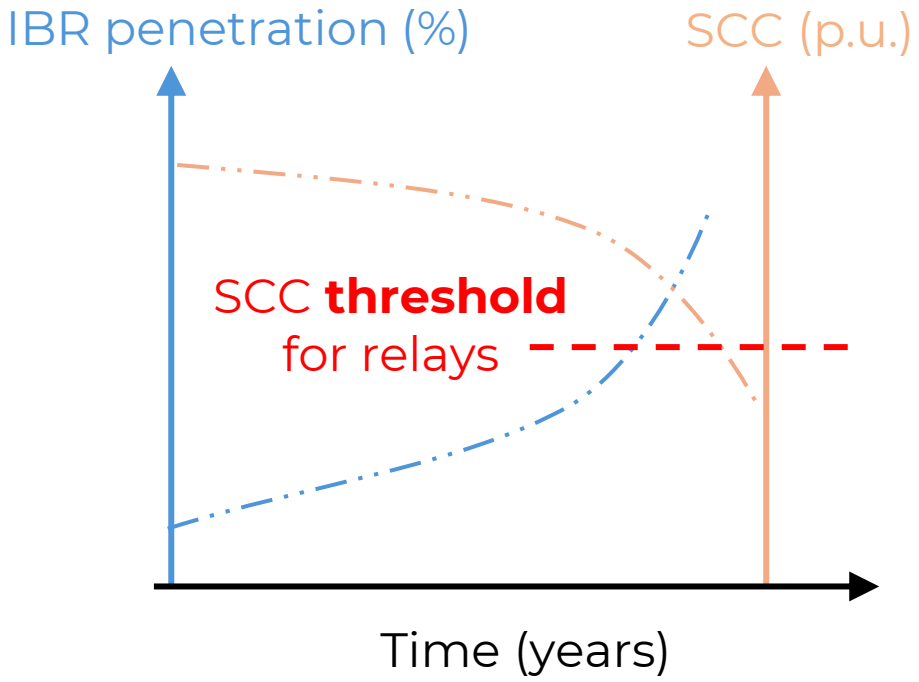


Electronic converters (**burning of transistors**)

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Electromechanical physics



How to tackle the decline in SCC?

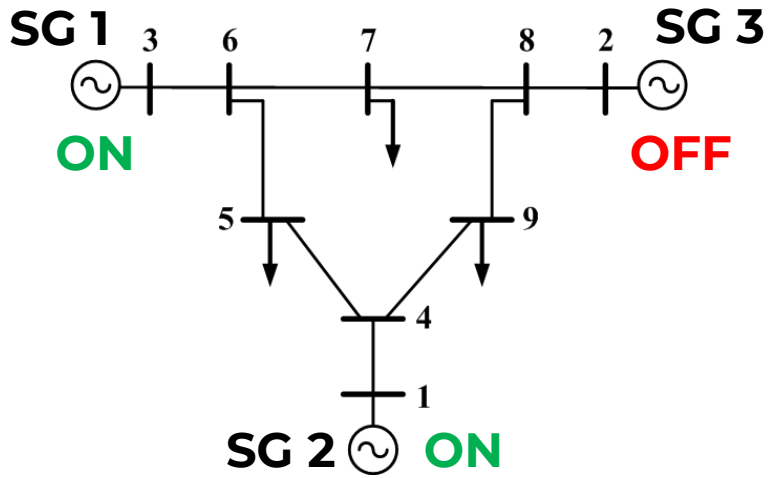
Option 1: invest in **grid-supporting assets**

Installing **synchronous condensers**



Option 2: security-constrained **generation dispatch**

Guarantee **sufficient generation online** for SCC



How to tackle the decline in SCC?

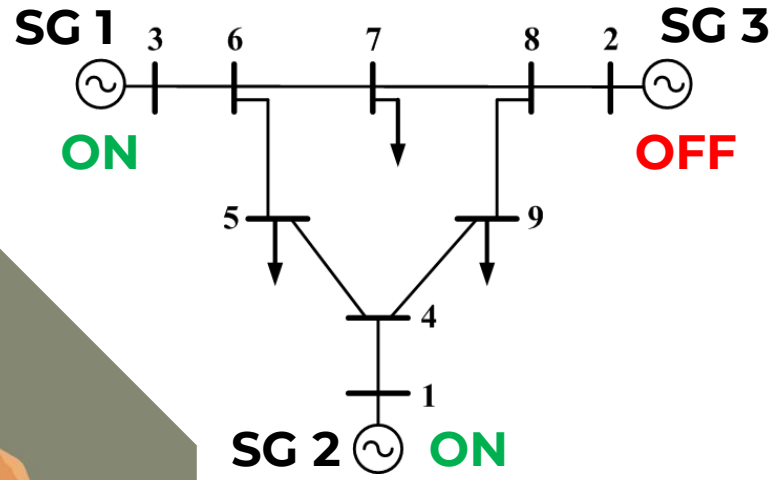
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Both options imply **economic cost**!

How to tackle the decline in SCC?

In this work, **we focus** on **option 2**

(although sync condensers could also be considered in our framework)

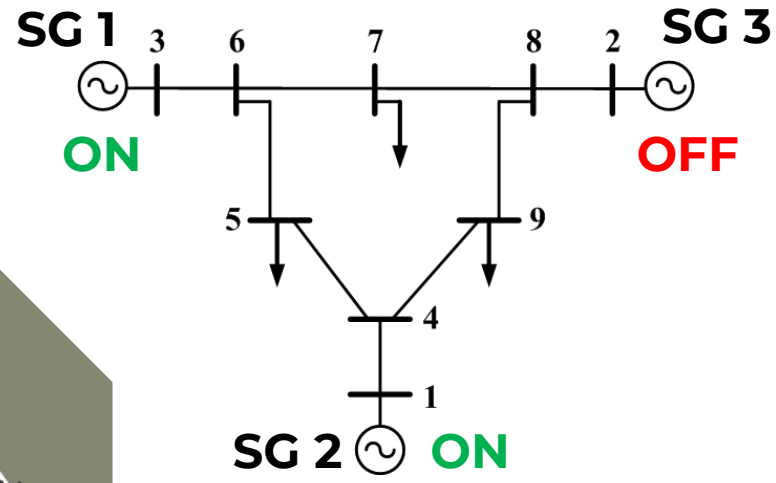
The **questions** we want to shed some light on are:

*Would it make sense **to create a market** for **SCC services**?*

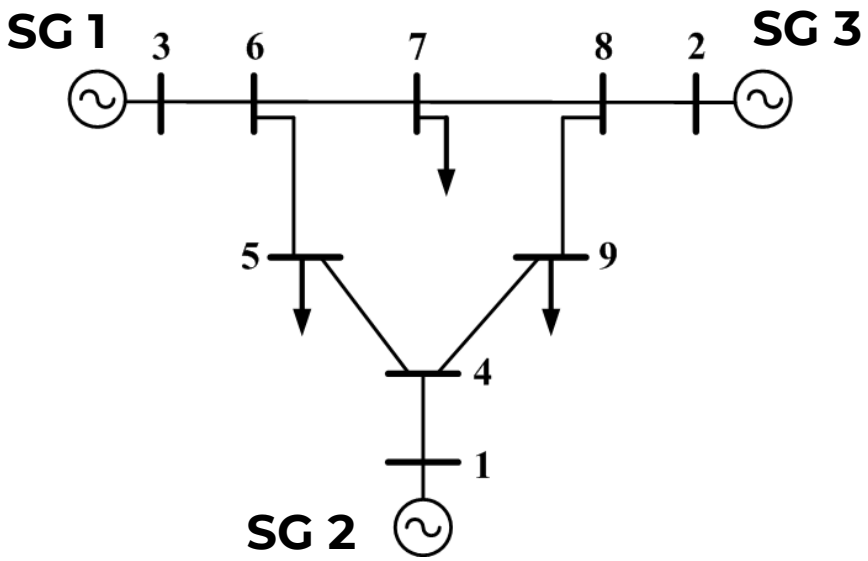
Would it be a **competitive market**, leading eventually to **lower costs for consumers**?



Guarantee **sufficient generation online** for SCC



Guaranteeing sufficient SCC: constraints in generation dispatch



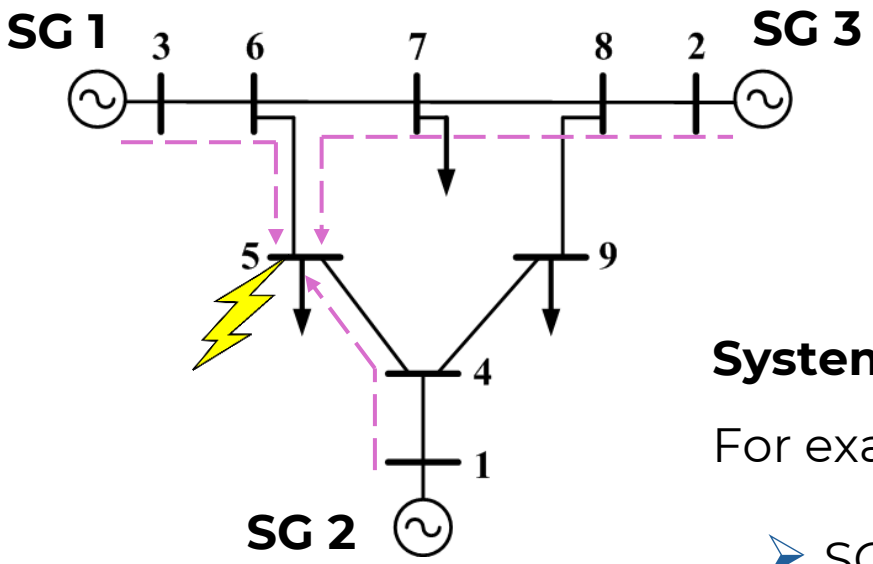
Guarantee at every bus that **SCC** is **above the threshold** [1]:

SCC from SGs SCC from IBR

$$I_{b_{\mathrm{SC}}} = \frac{\sum_{g\in\mathcal{G}} Z_{b\Psi(g)} \mathrm{I}_g u_g + \sum_{c\in\mathcal{C}} Z_{b\Phi(c)} \mathrm{I}_c \alpha_c}{Z_{bb}} \geq \mathrm{I}_{b_{\mathrm{lim}}}$$

[1] Z. Chu and F. Teng, "Short circuit current constrained UC in high IBG-penetrated power systems," *IEEE Trans. Power Syst.*, 2021

Guaranteeing sufficient SCC: constraints in generation dispatch



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System impedance (Z) leads to different SCC levels for each bus

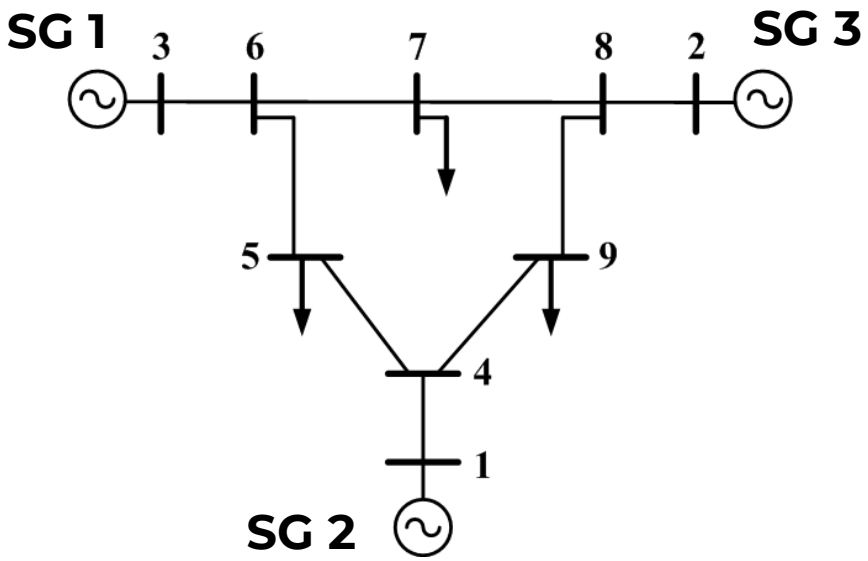
For example:

- SG 1 and SG 2 would contribute a larger share of SCC to bus 5, due to a closer **electrical distance**

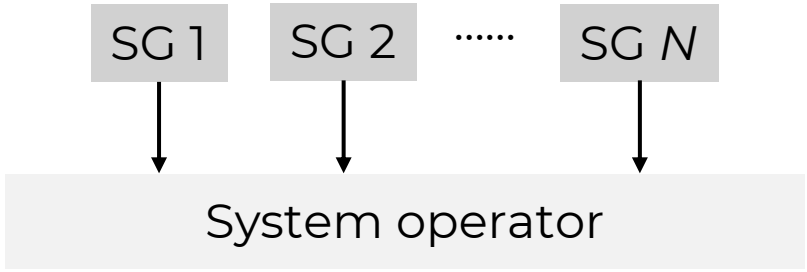
SCC provision is **very locational**

[1] Z. Chu and F. Teng, "Short circuit current constrained UC in high IBG-penetrated power systems," *IEEE Trans. Power Syst.*, 2021

Potential problem: market power in SCC services?



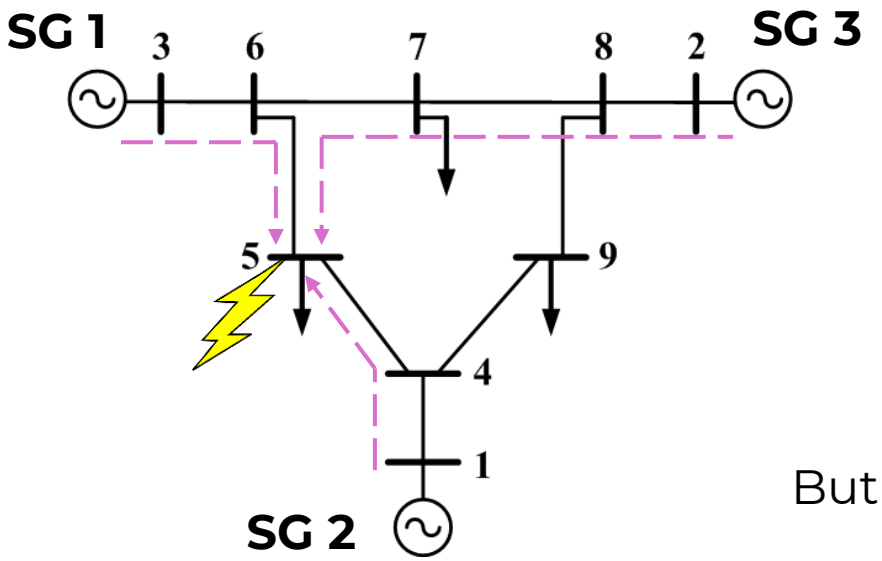
SCC may be procured as an **ancillary service** [2]



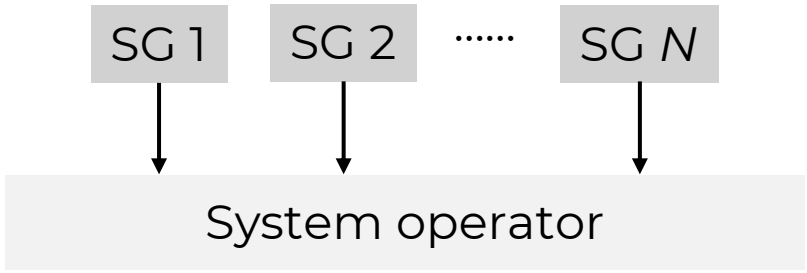
Receives offers and clears energy and SCC markets

[2] Z. Chu, J. Wu and F. Teng, "Pricing of short circuit current in high IBR-penetrated system," *Electric Power Syst. Research*, 2024

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SCC may be procured as an **ancillary service** [2]



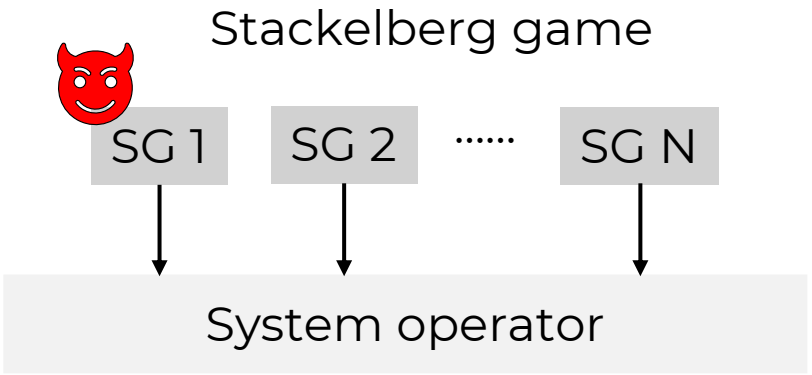
Receives offers and clears energy and SCC markets

But SCC provision is **very locational**...

May an agent at an **advantageous location** exert market power to **distort SCC prices**?

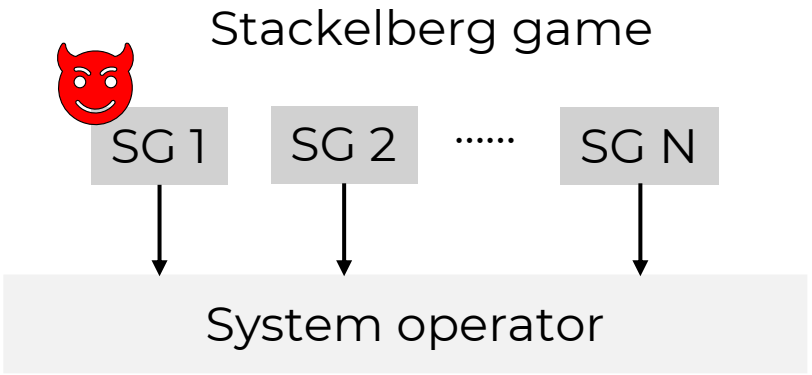


How to model imperfect competition in SCC services?



- Self-interested SG 1 → ***strategically*** bids for profits
- Other SGs → assumed to bid truthfully
- System operator → clears market

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Bilevel optimization model

Upper level (**SG 1**)
- Objective: **max** profit
- Actions: ***dishonest*** energy and SCC bids

Lower level (System operator)
- Objective: **min** system operating cost
- Actions: receive bids from upper level, dispatch generators and define market prices

Solving a bilevel optimization problem

Traditional way to solve a ***convex*** bilevel problem:

Upper level
- objective: max (·)
- a set of constraints

Lower level
- objective: min (·)
- a set of constraints

KKT conditions [3]



only for lower level

Upper level
- objective: max (·)
- a set of constraints

Lower level
- replaced by a set of **equivalent constraints** (KKT conditions)

[3] A. Conejo and C. Ruiz, "Complementarity, not optimization, is the language of markets," *IEEE Journal of Power & Energy*, 2020

Solving a bilevel optimization problem

Traditional way to solve a ***convex*** bilevel problem:

- Upper level
  - objective: max (·)
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- Lower level
  - objective: min (·)
  - a set of constraints

KKT conditions [3]



only for lower level

Equivalent **single-level** optimization

- Upper level
  - objective: max (·)
  - a set of constraints
  - 
  - set of constraints corresponding to the **Lower level**

[3] A. Conejo and C. Ruiz, "Complementarity, not optimization, is the language of markets," *IEEE Journal of Power & Energy*, 2020

Challenge: bilevel optimization + binary variables

Unfortunately, **our bilevel model** cannot be easily solved

- SCC from SGs is inherently related to **unit commitment** (binary variables) → discrete **non-convexities**

{0, 1} (gen state OFF or ON)

$$I_{b_{\mathrm{SC}}} = \frac{\sum_{g \in \mathcal{G}} Z_{b\Psi(g)} \mathrm{I}_g u_g + \sum_{c \in \mathcal{C}} Z_{b\Phi(c)} \mathrm{I}_c \alpha_c}{Z_{bb}} \geq \mathrm{I}_{b_{\mathrm{lim}}}$$

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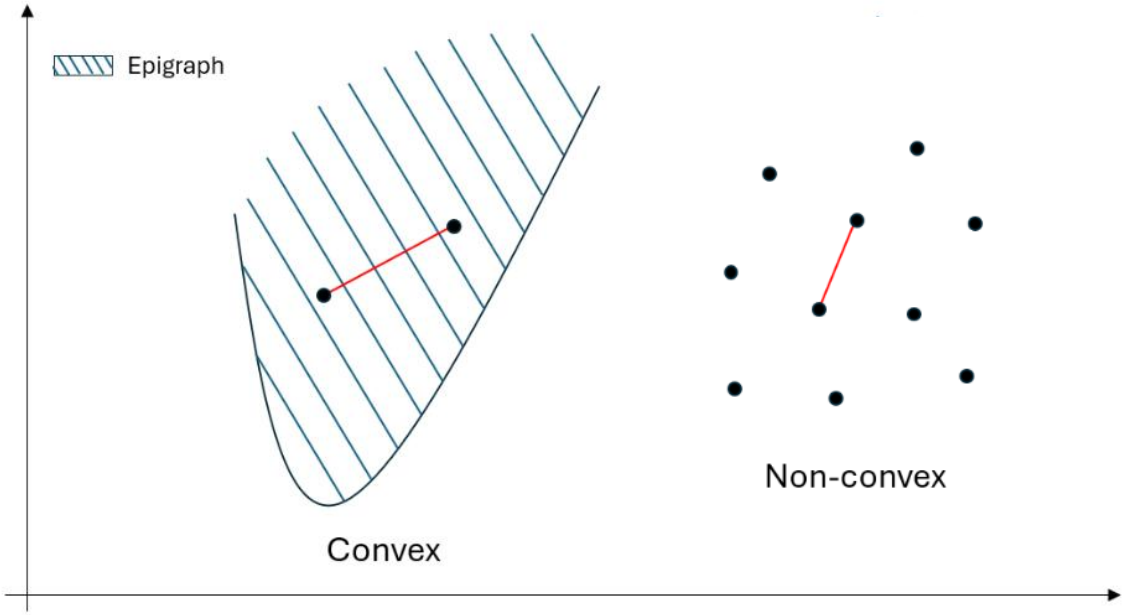
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If you want to read more:

L. Badesa, "Duality in optimization, KKT and shadow prices – With applications to electricity markets," 2025



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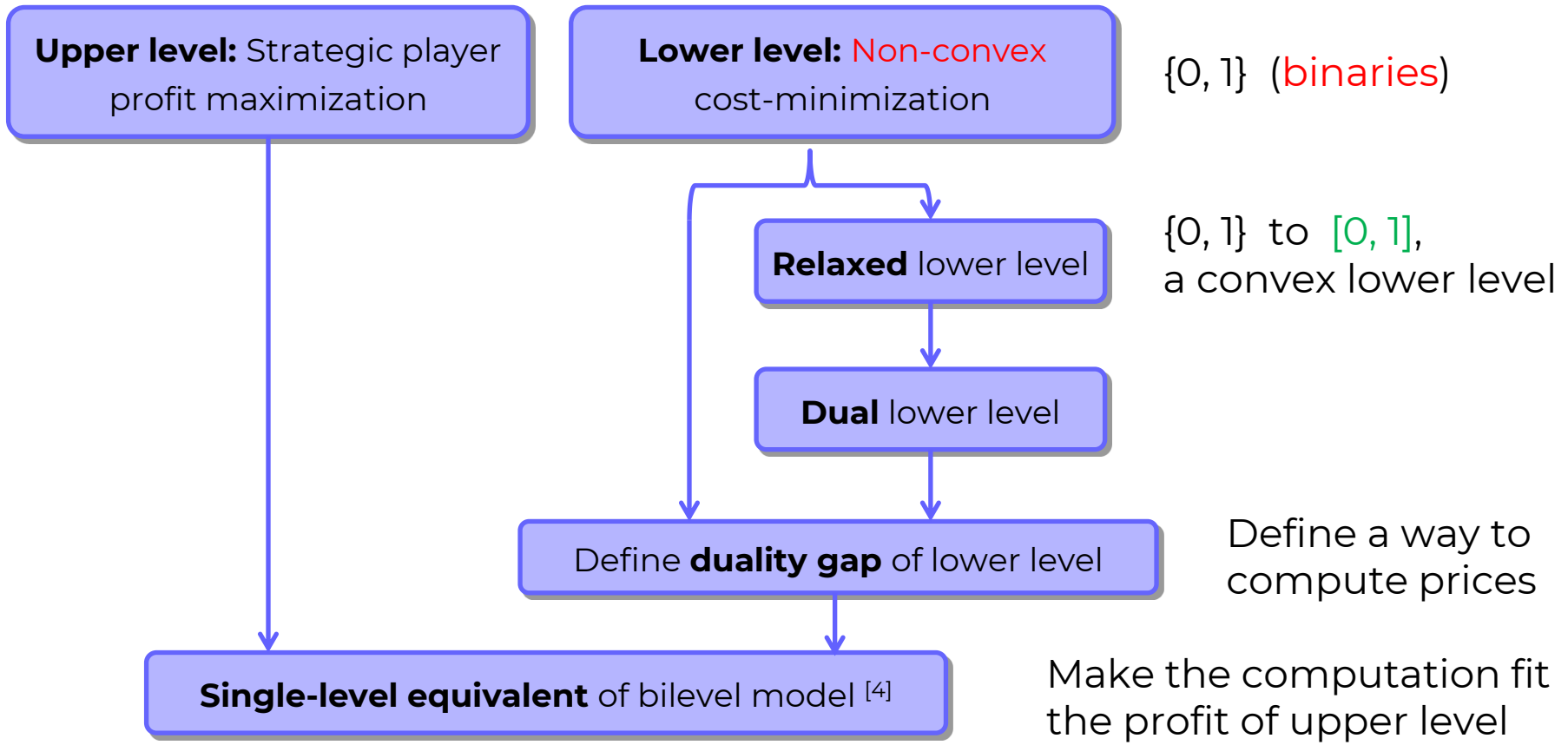
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The traditional solution method **requires** the **lower level** to be **convex**

- Otherwise, the KKT conditions are not an equivalent representation (they are necessary but not sufficient conditions for optimality)

How to address non-convexity?

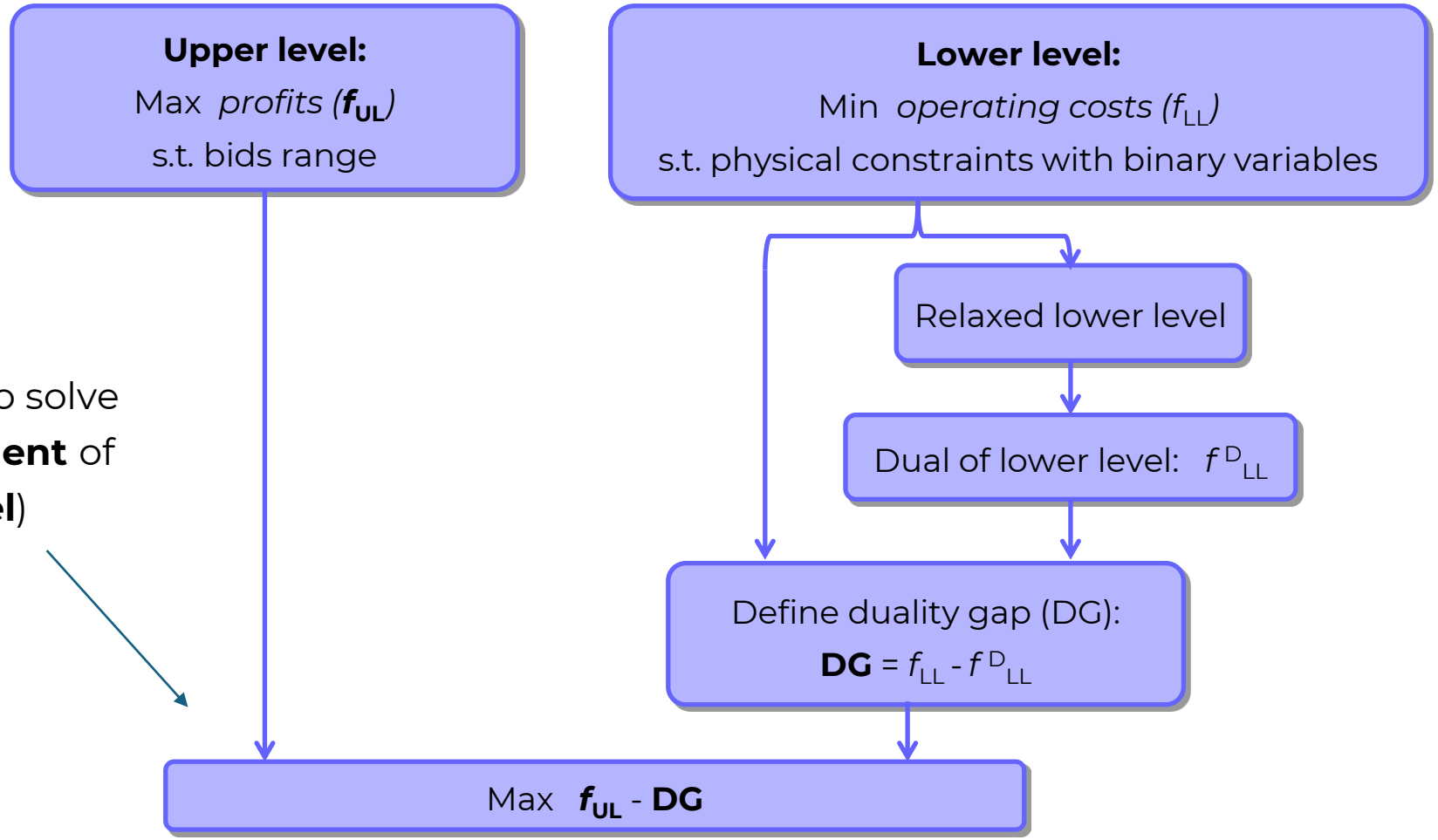
Solution:
primal-dual formulation
for lower level



[4] Y. Ye *et al.*, "Incorporating Non-Convex Operating Characteristics into Bi-Level Optimization Electricity Market Models," *IEEE Trans. on Power Syst.*, 2020

How to address non-convexity?

(clarification on how to solve a **single-level equivalent** of the **bilevel model**)

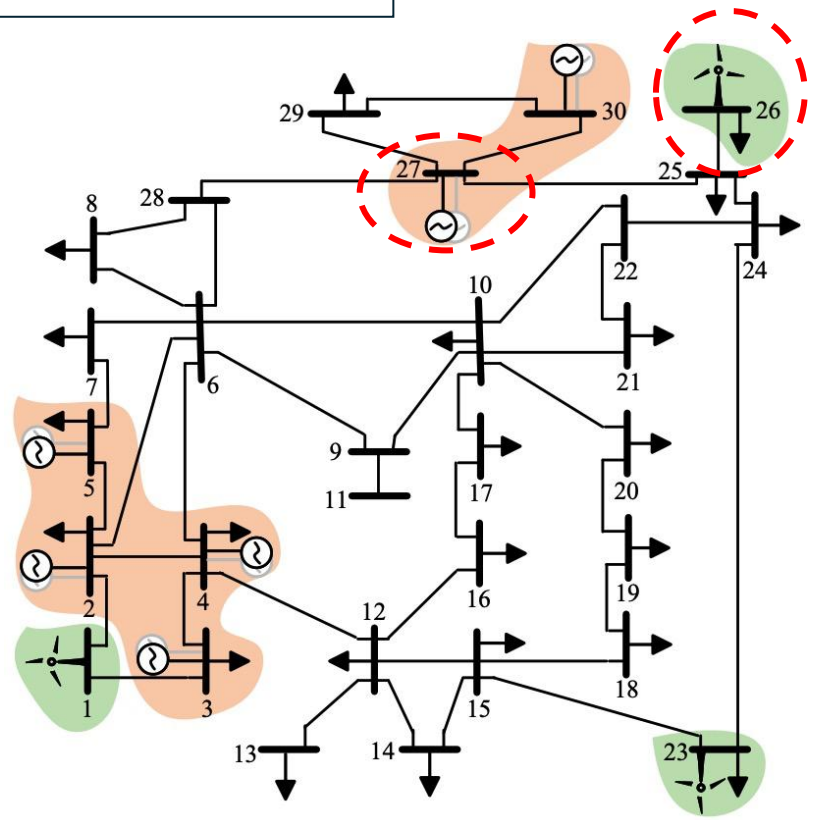


Research results — roots of market power

Locality of SCC + Discrete supply mechanism + Hard security requirement

The SCC at the system's **weak bus 26**,
is highly dependent on SGs at bus 27

- Contribution from **SGs at bus 27**: **3.34 p.u.**
- Contribution from **all other SGs**: 2.77 p.u. at maximum



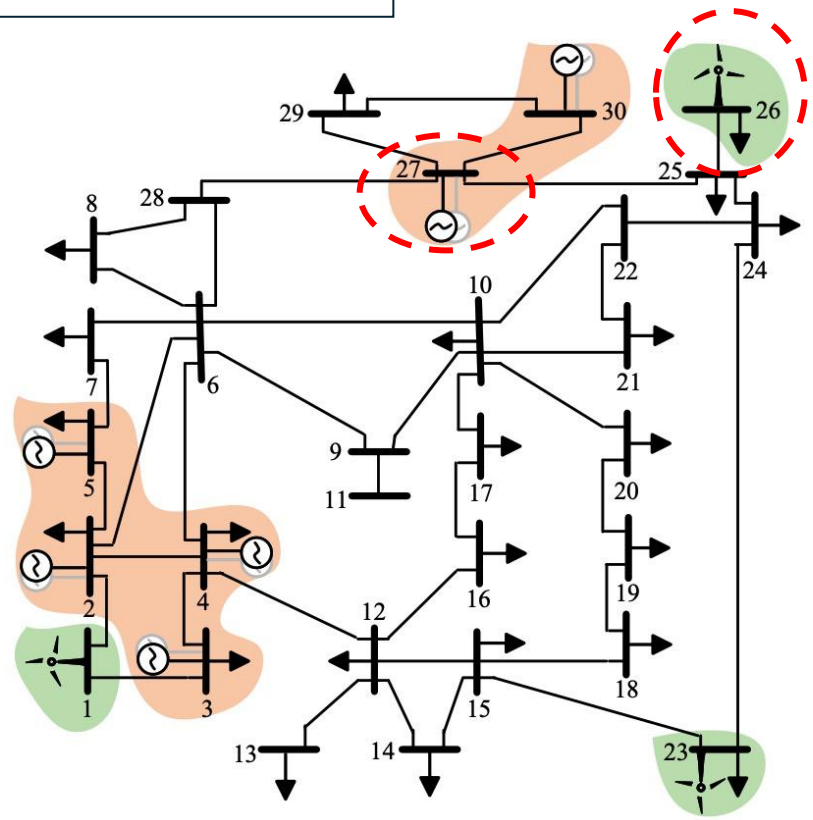
Modified IEEE 30-bus system

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Modified IEEE 30-bus system

- The ability to support SCC is determined by **electrical distance**
- Only *nearby* SGs can provide *sufficient SCC*
- While distant SGs have no substitutable value, creating a **natural local supply barrier**

Research results — roots of market power

Locality of SCC + **Discrete supply mechanism** + **Hard security requirement**

$$I_{b_{\mathrm{SC}}} = \frac{\sum_{g \in \mathcal{G}} Z_{b\Psi(g)} \mathrm{I}_g u_g + \sum_{c \in \mathcal{C}} Z_{b\Phi(c)} \mathrm{I}_c \alpha_c}{Z_{bb}} \geq \mathrm{I}_{b_{\mathrm{lim}}}$$

- The **SCC provision** of SGs **is discrete**: it's an **all-or-nothing service**
  (an online generator provides a given volume of SCC; offline provides none)
- A bus must meet a minimum **SCC threshold**, otherwise grid security is threatened

Research results — consequences of market power

	Bidding **truthfully**	Bidding **strategically**
SCC revenue of **SG**s at **bus 27**	44.4 k€	147.9 k€ (**+3.3x**)
Consumers cost	65.0 k€	175.6 k€ (**+2.7x**)

Key takeaways

Would it make sense ***to create a market*** *for* ***SCC services****?*

Maybe, but **with care**

Certain generators could more than

🚨 **triple their revenues** 🚨

from this service if **market power** issues are not properly mitigated

- SCC ancillary service is **highly *location-dependent***: generators near vulnerable buses possess inherent market power due to *irreplaceable SCC*

Key takeaways

Would it make sense ***to create a market*** *for* ***SCC services****?*

Maybe, but **with care**

(**market power** issues)

So ***what can we do*** *about it?*

Further work needed to properly answer this question

- Installing **synchronous compensators owned by the TSO** in the most critical buses could help make the remaining SCC market more competitive
- **Price cap**: directly limit SCC service prices

Future work

- Study interactions between **multiple strategic players**

  (**EPEC**, Equilibrium Problem with Equilibrium Constraints)

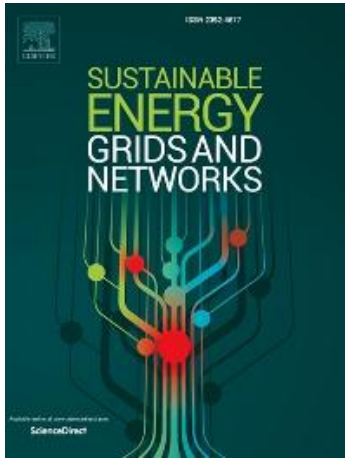
- Consider **unbalanced faults** (current work only considers three-phase faults)

- Analyze markets for **other stability services**, such as voltage stability

Paper reference

P. Wang and L. Badesa, "Imperfect Competition in Markets for Short-Circuit Current Services,"

Sustainable Energy, Grids and Networks, vol. 47, 2026





Peng Wang

(**first author**, PhD student at UPM)

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